Dynamic network analysis of protein structural change

Aydin Wells[1], Siyu Yang[1], Khalique Newaz[2], and Tijana Milenković[1]

[1] **Department of Computer Science and Engineering; University of Notre Dame**

[2] Department of Computational Systems Biology; University of Hamburg

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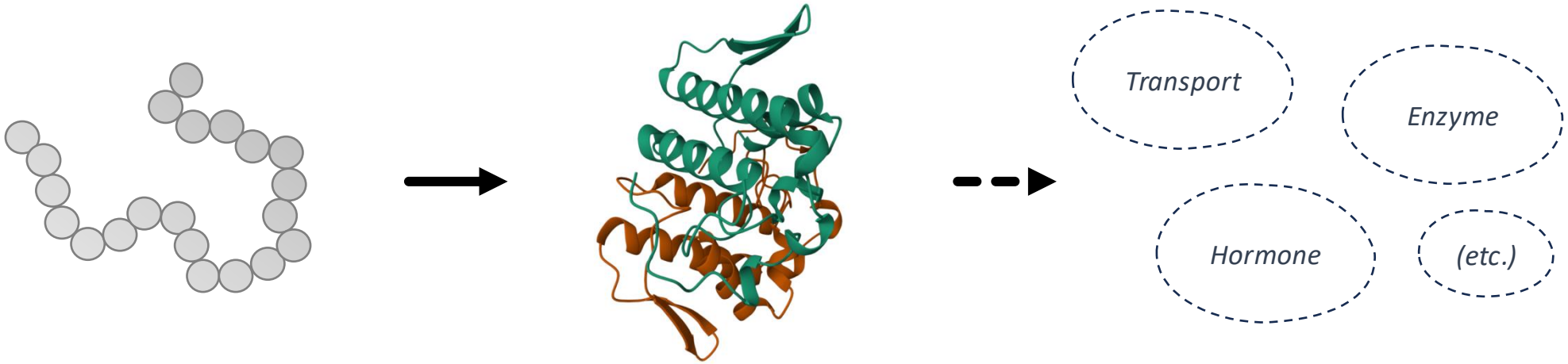
Introduction and previous work

- A protein's amino acid sequence folds into a 3D structure
- A protein's 3D structure directs what and how it interacts with other biomolecules to carry out cellular roles
- Analysis of protein structures is critical for understanding how proteins operate



Amino acid sequence **Protein 3D structure** **Types of cellular roles**

Introduction and previous work

- A protein's amino acid sequence folds into a 3D structure
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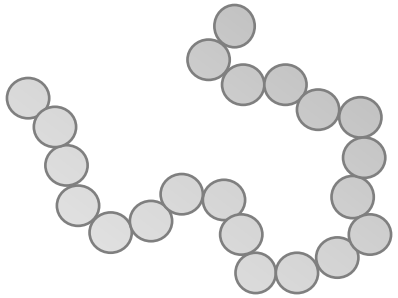
- Analysis of protein structures is critical for understanding how proteins operate

✗ *Experimental* ✓ *Computational*



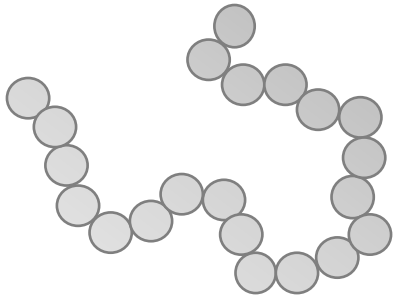
Amino acid sequence

Protein 3D structure

Roles of many proteins remain unknown

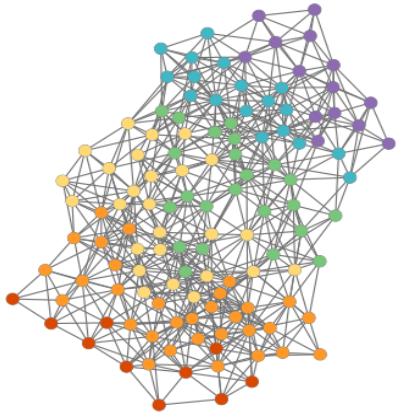
Introduction and previous work

- Protein sequence or 3D structural features
- Protein structure networks (PSNs)



Amino acid sequence

Protein 3D structure



Protein structure network

Introduction and previous work

- Protein sequence or 3D structural features
- Protein structure networks (PSNs)
- We and others have shown that PSN approaches are effective in the task of protein structure classification (PSC)

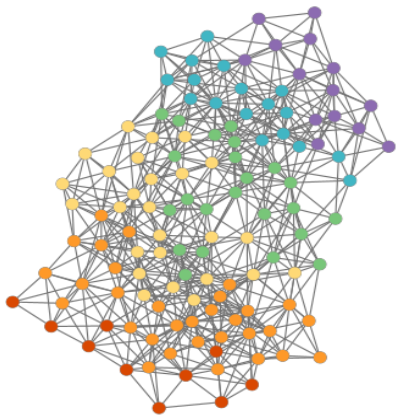
In the task of PSC, PSN approaches are superior to:

Sequence approaches	3D structural approaches
DAAComposition	TM-Align
AAComposition	DaliLite
SVMfold	GIT

while having equally, or better:

Accuracy Run time

Newaz et. al., 2020, 10.1098/rsos.191461



Protein structure network

Introduction and previous work

- Traditional PSN methods (including our previous approaches) model the whole (native) 3D structure of a protein as a static PSN

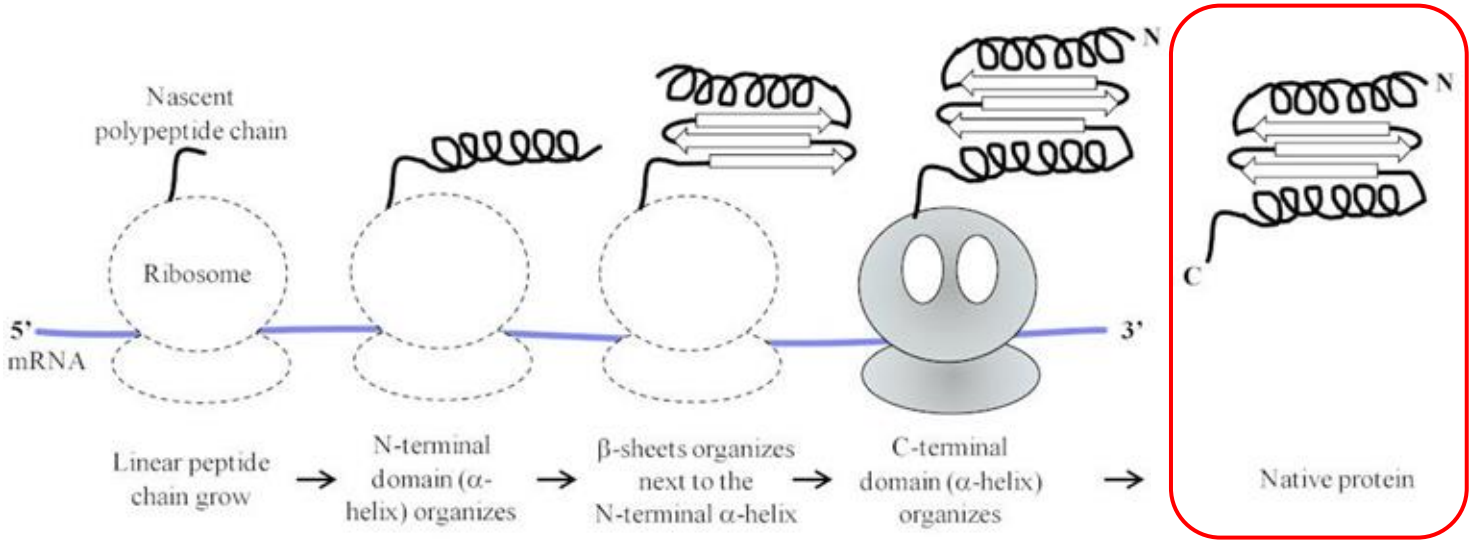


Figure from Biyani et al., 2012, 10.5772/53329

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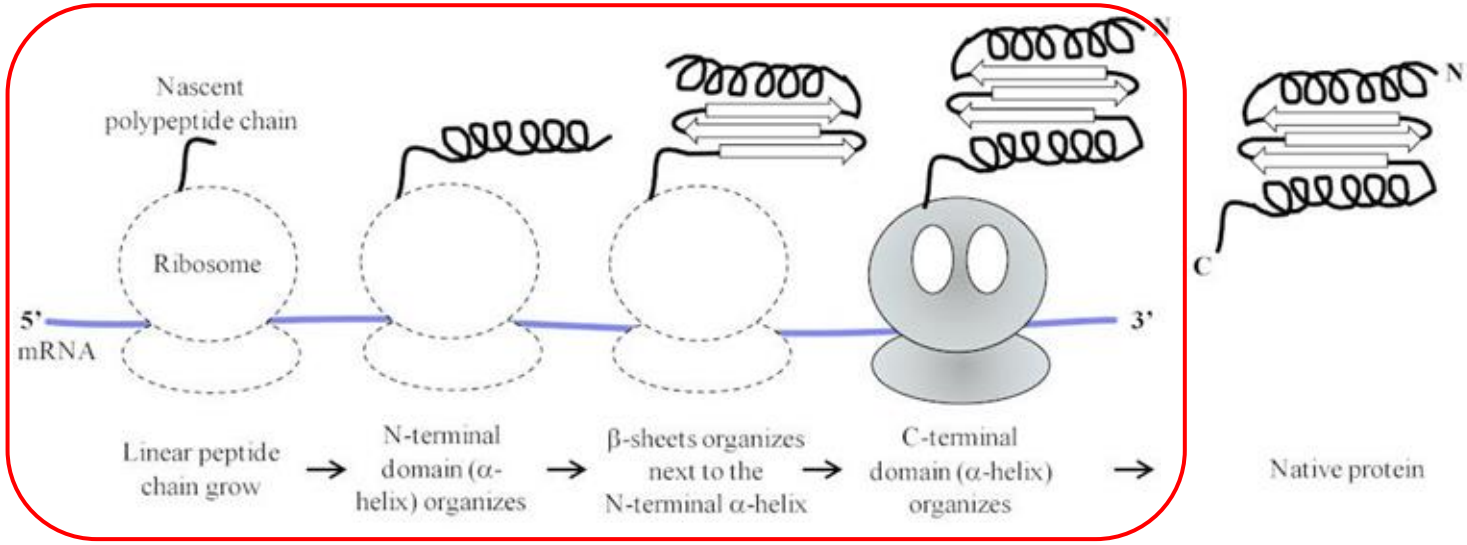

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- Few kinds of experimental methods that can obtain such data, and such methods are not scalable
- Experimental data (on both large- and small- scale) on protein 3D sub-structures are lacking

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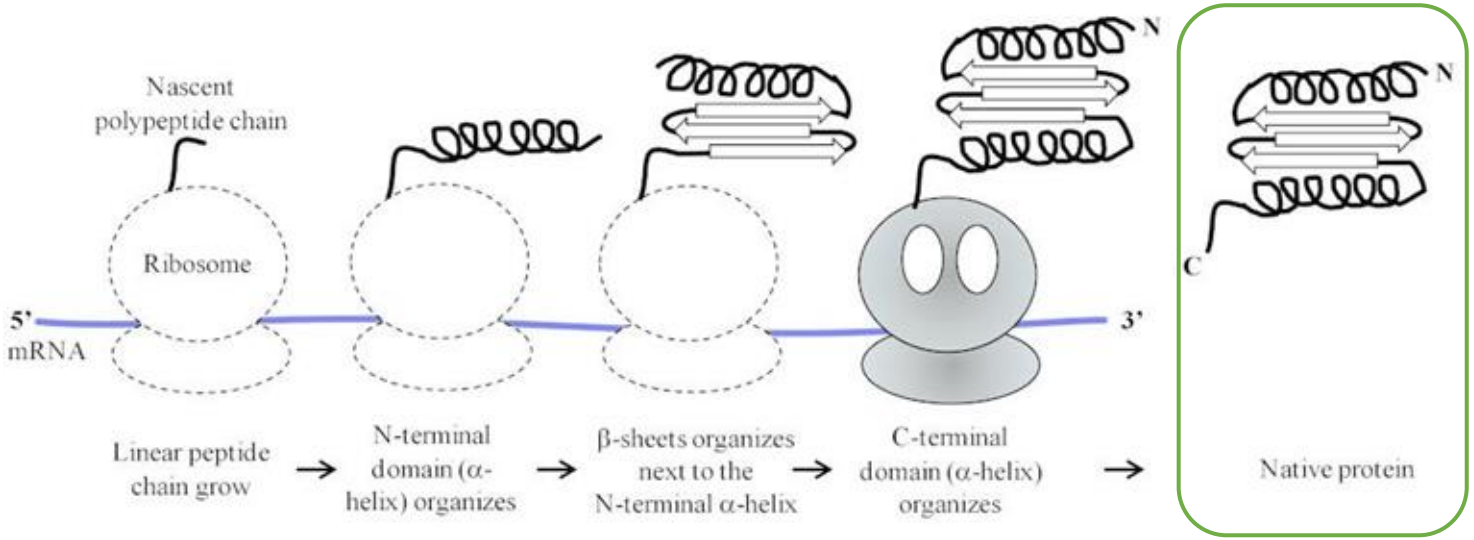

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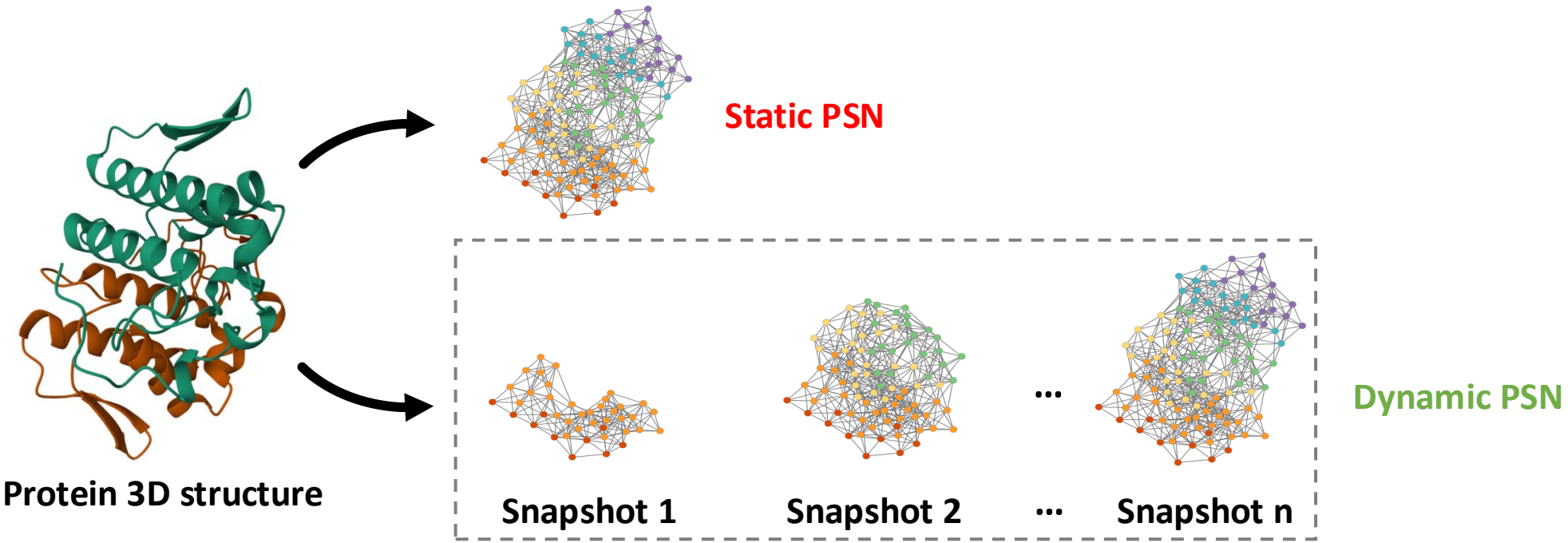
- Few kinds of experimental methods that can obtain such data, and such methods are not scalable
- Experimental data (on both large- and small- scale) on protein 3D sub-structures are lacking
- As a proxy, our lab constructed a new PSN type using the native 3D structures of proteins

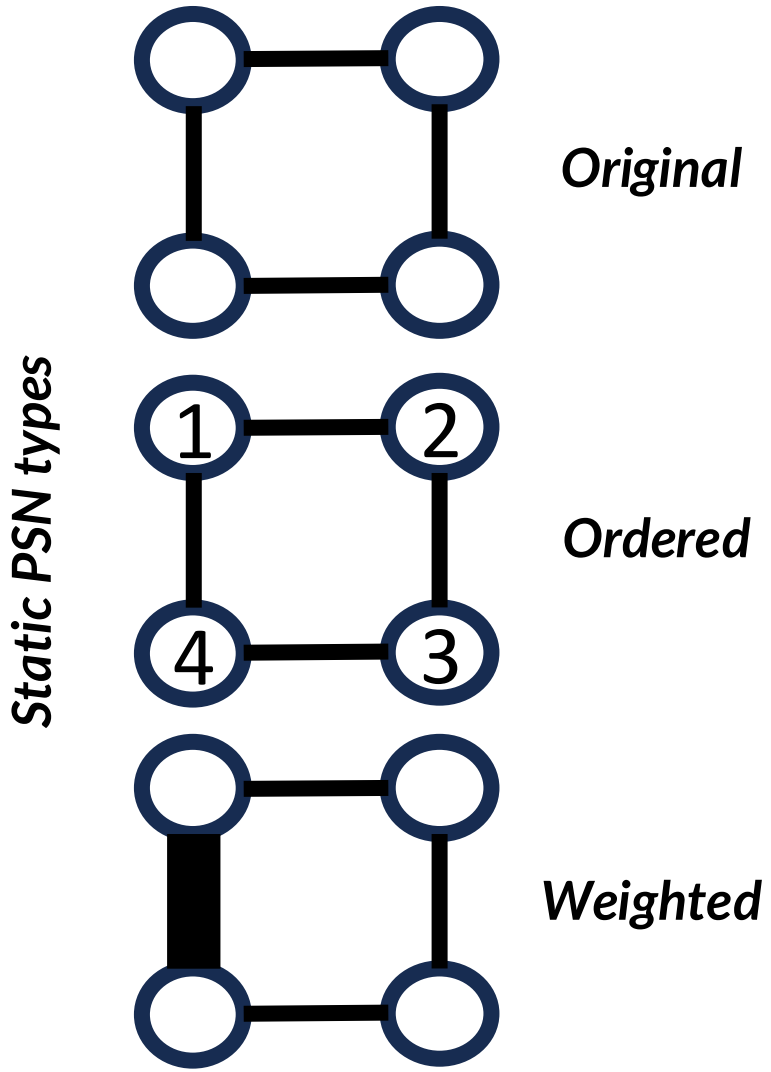
Introduction and previous work

- Traditional PSN methods (including our previous approaches) model the whole (native) 3D structure of a protein as a static PSN
- A new PSN type to capture dynamic 3D structural information – dynamic PSN



Introduction and previous work

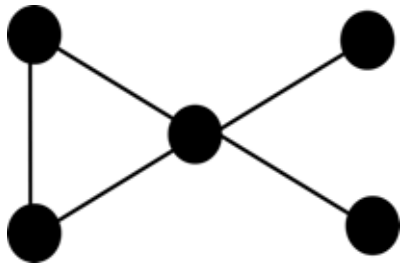
- 72 datasets containing ~44,000 CATH/SCOPe protein domains
- Compared dynamic PSN to three static PSN types
- Protein structure classification task (PSC)

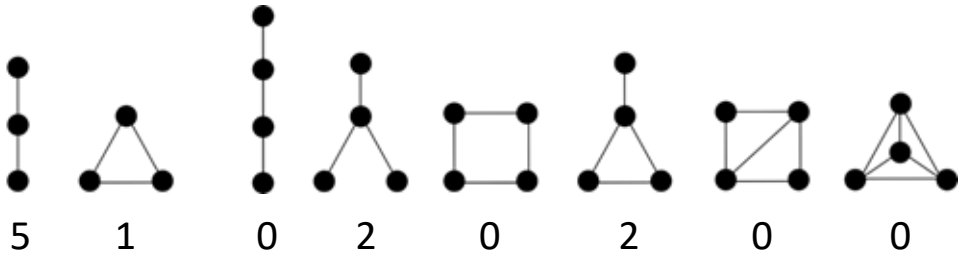


Introduction and previous work

- 72 datasets containing ~44,000 CATH/SCOPe protein domains
- Compared dynamic PSN to three static PSN types
- Protein structure classification task (PSC)
- Network-based features (i.e., graphlets)
- Standard 5-fold cross validation
- Logistic regression classifier

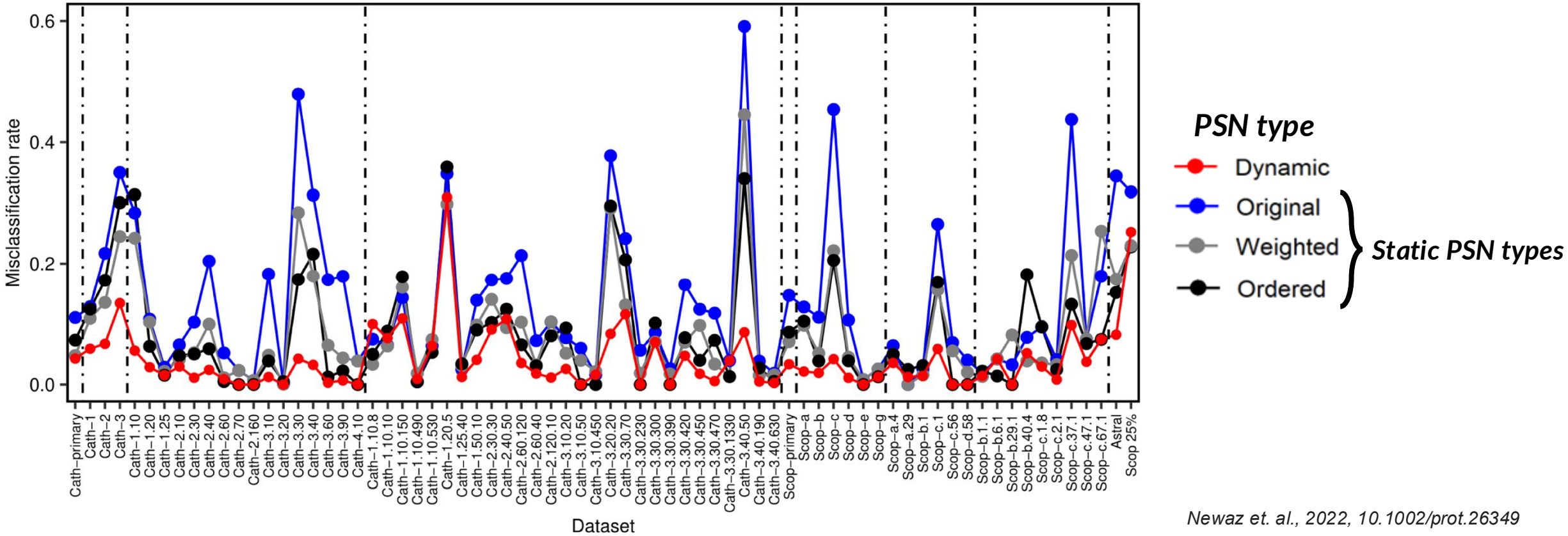
Toy PSN:





(Original) graphlet features

Introduction and previous work



Newaz et. al., 2022, 10.1002/prot.26349

PSN type	# where dynamic is **superior**	# where dynamic is **tied**	# where dynamic is **inferior**	**Adjusted *p*-value**
Original	**64** / 72	**4** / 72	**4** / 72	$\leq 10^{-6}$
Weighted	**61** / 72	**2** / 72	**9** / 72	$\leq 10^{-6}$
Ordered	**51** / 72	**13** / 72	**8** / 72	$\leq 10^{-6}$

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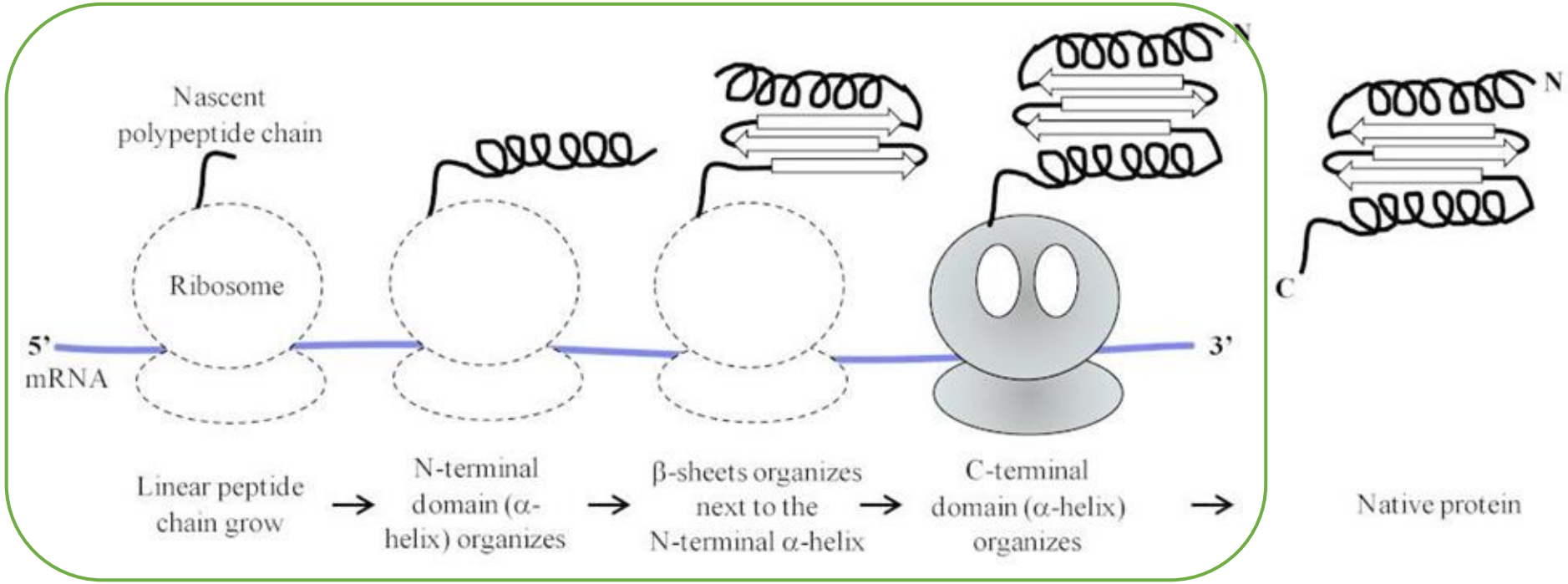
Motivation and contribution

- Experimental data on protein 3D sub-structures (dynamic data) is lacking



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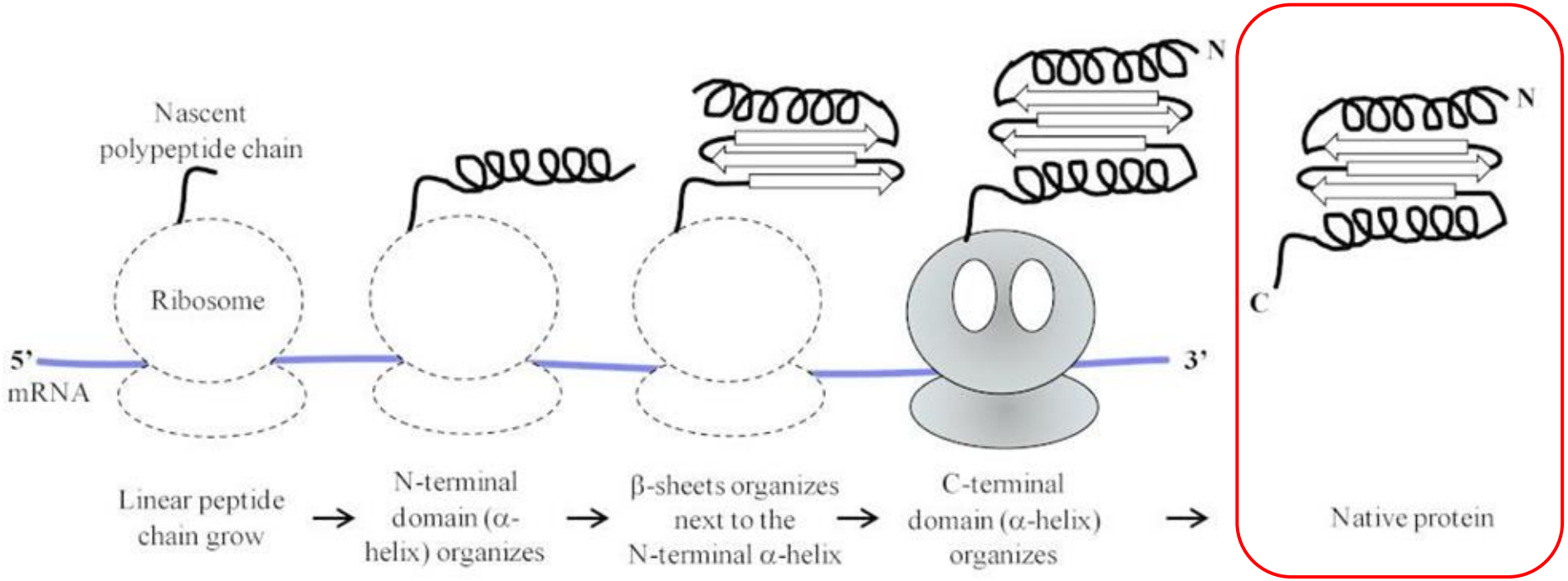
- A dynamic model (i.e., dynamic PSNs) on static data (i.e., native 3D structures)



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Motivation and contribution

- Experimental data on protein 3D sub-structures (dynamic data) is lacking
- A dynamic model (i.e., dynamic PSNs) on static data (i.e., native 3D structures)

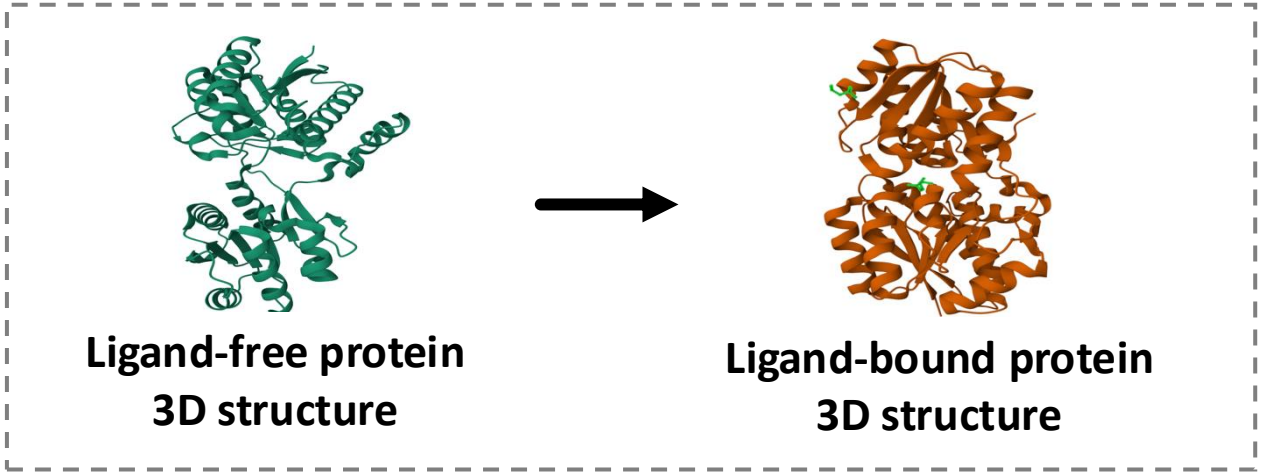
Will capturing dynamic 3D sub-structural data – rather than "just" the native structure – with dynamic PSNs yield further improvements?

Motivation and contribution

- Experimental data on protein 3D sub-structures (dynamic data) is lacking
- A dynamic model (i.e., dynamic PSNs) on static data (i.e., native 3D structures)

Will capturing dynamic 3D sub-structural data – rather than "just" the native structure – with dynamic PSNs yield further improvements?

- A proxy for dynamic data (i.e., protein structural change)



Protein-ligand pair

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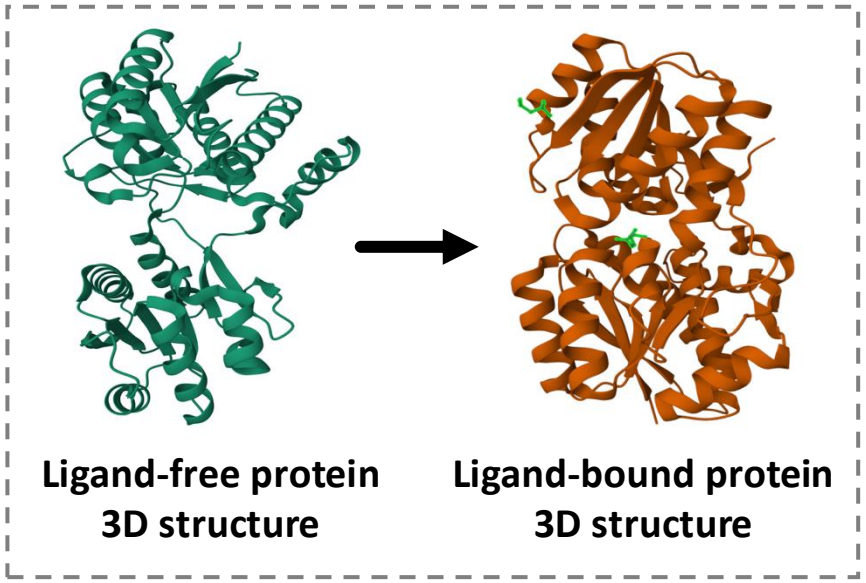
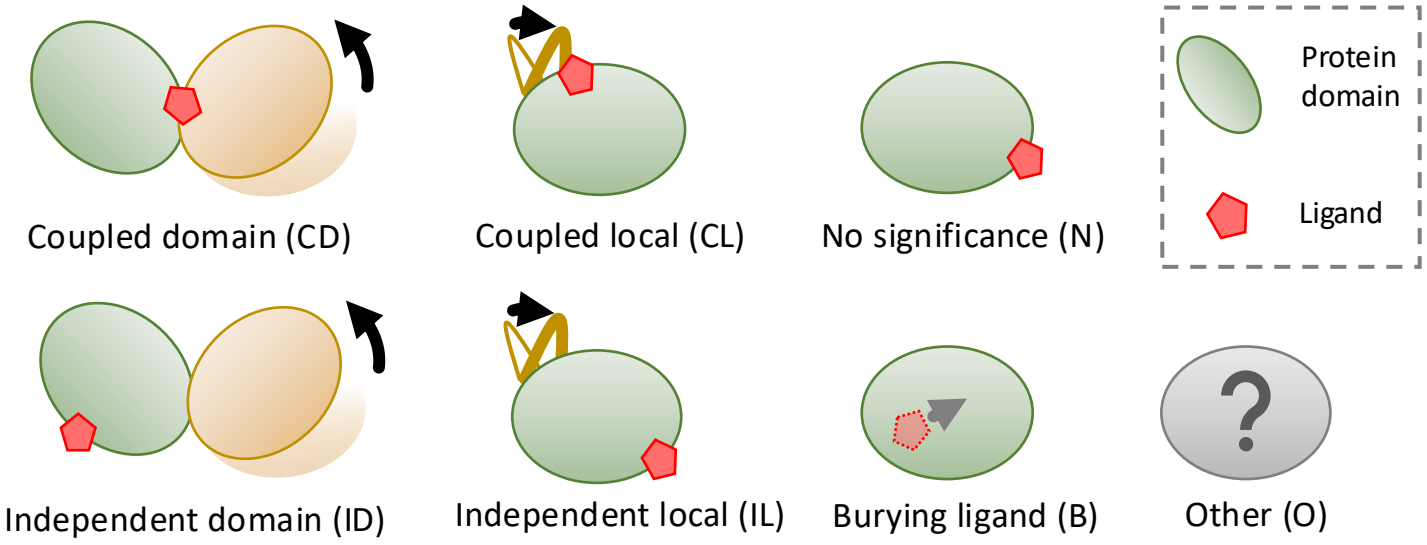
Methods and evaluation

- Protein Structure Change DataBase (PSCDB) → 1,084 protein-ligand pairs)
- Ligand-bound protein motion (7 motion types)
- Cleaned and consolidated data (i.e., data quality thresholds / datasets)



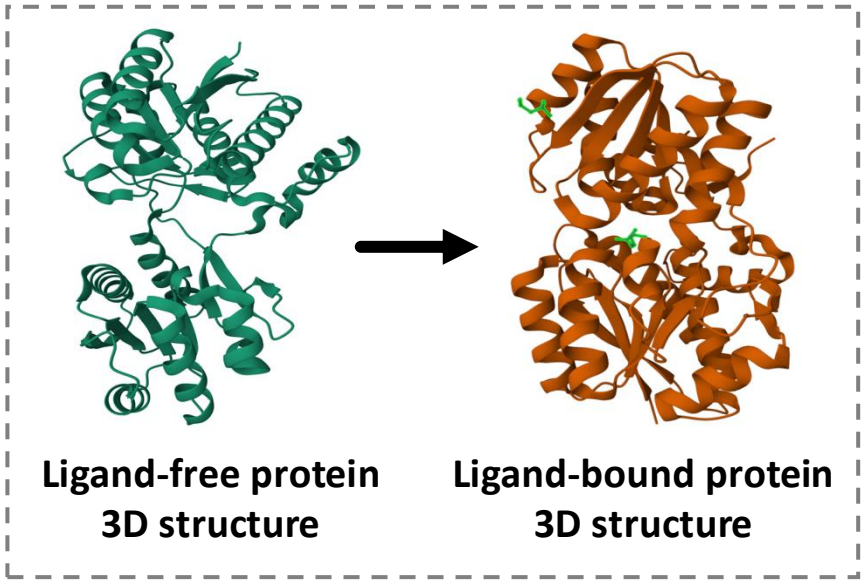
Protein-ligand pair



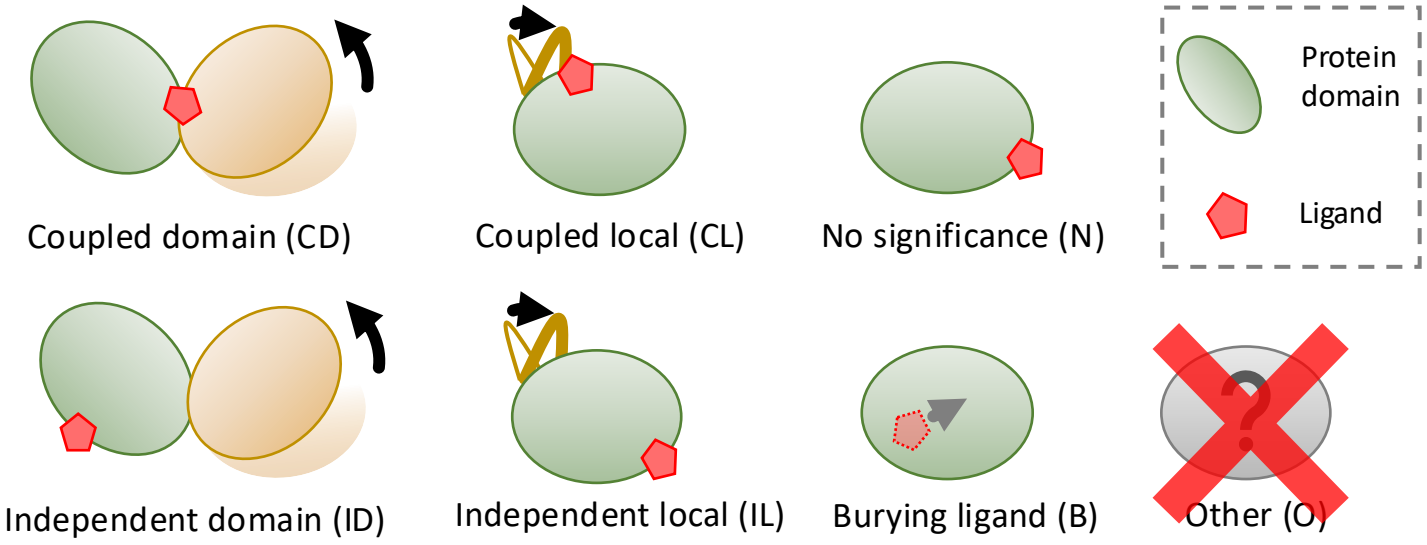
Protein-ligand motion types

Methods and evaluation

- Protein Structure Change DataBase (PSCDB) → 1,084 protein-ligand pairs)
- Ligand-bound protein motion (6 motion types)
- Cleaned and consolidated data (i.e., data quality thresholds / datasets)



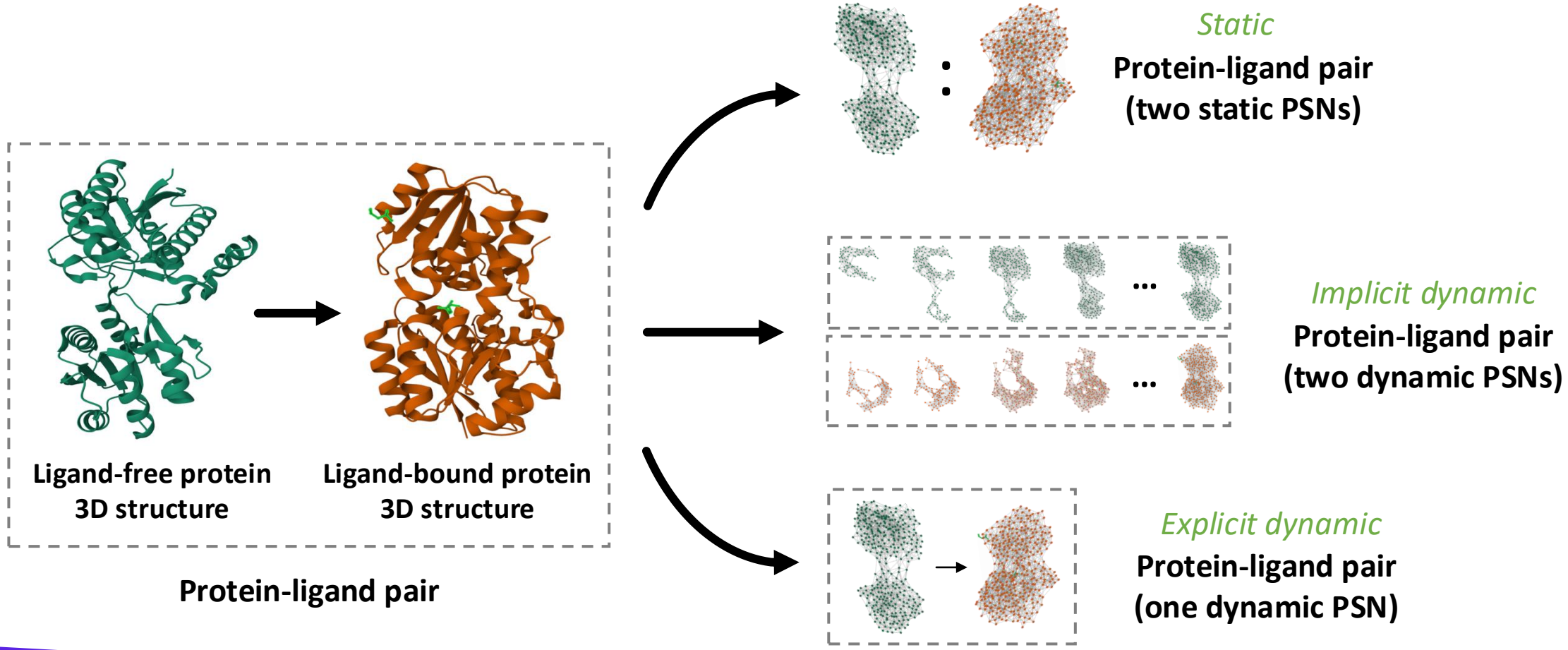
Protein-ligand pair



Protein-ligand motion types

Methods and evaluation

- For each protein 3D structure before (free) and after (bound) ligand binding, we model them as both static and dynamic PSNs



Methods and evaluation

- For each PSN type, we extract graphlet features
- Cross-fold validation
- Multi-label classification method (w/ 6 labels and 7 datasets)

$$\boldsymbol{For\ each\ dataset} = \frac{\#\ of\ protein{-}ligand\ pairs\ classified\ correctly}{Total\ \#\ of protein{-}ligand\ pairs\ in\ a\ dataset}$$

- (We compare our results to basic benchmarks (i.e., majority class, TM-score & RMSD))

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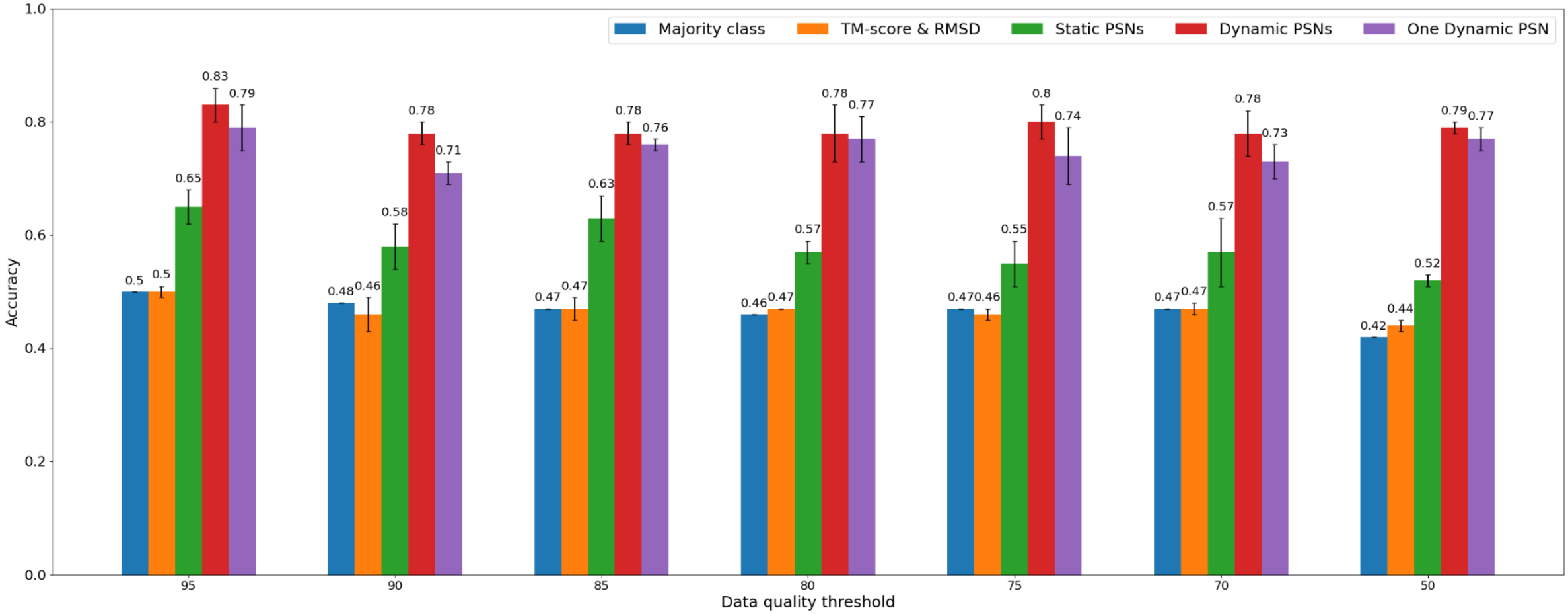


Results and conclusion

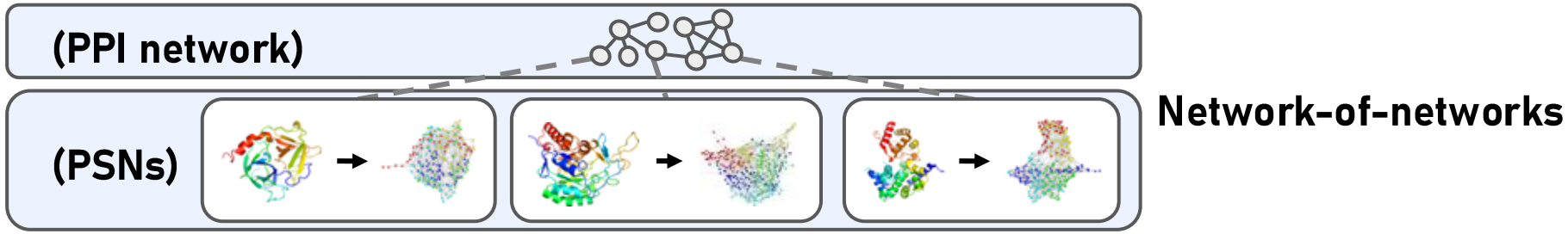
Dynamic PSN features extracted from protein-ligand motion data more accurately classify and explain the six different types of protein structural changes than static PSN features (and basic benchmarks).

- Dynamic PSNs are generalizable to dynamic data (dynamic model on dynamic data)
- Improve on dynamic PSNs (w/ experimental 3D structures of protein fold intermediates, and on large scale)

Results and conclusion

Dynamic PSN features extracted from protein-ligand motion data more accurately classify and explain the six different types of protein structural changes than static PSN features (and basic benchmarks).

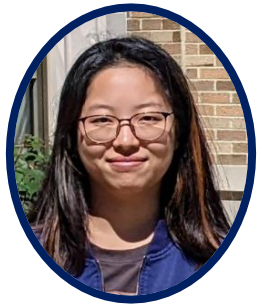
- Dynamic PSNs are generalizable to dynamic data (dynamic model on dynamic data)
- Improve on dynamic PSNs (w/ experimental 3D structures of protein fold intermediates, and on large scale)
- Integrating dynamic PSNs with protein-protein interactions (PPI) data for protein function predictions (Gu et al., Bioinformatics, 2022)



Thank you!

Professor Tijana Milenković

Siyu Yang



Jennefer Morones

Aydin Wells
awells24(at)nd(dot)edu
ISMB2024 – 3DSIG
07/12-16/2024

UNIVERSITY OF NOTRE DAME

Lucy Family Institute for Data & Society